

# Progress Report: SQuAD Sentence Saliency

## Discourse Priors & Class Balancing Experiments

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### 1. Summary of Objectives

This project evaluated sentence saliency detection (identifying answer-bearing sentences in SQuAD passages) to optimize long-context pruning and Question Generation. We evaluated 13 pointwise configurations across 5 balancing methods.

### 2. Silver Label Cleaning & Quality Audit

- Token-Level Intersection Filter:** To eliminate False Positives caused by trailing spaces or punctuation spilling over boundaries, we required that the character intersection of the sentence and the SQuAD answer text contains at least one non-stopword, alphanumeric token.
- Audit Agreement:** The local LLM-as-a-judge audit (using *Qwen2.5-1.5B-Instruct*) achieved an **82.00% agreement rate** (Cohen's Kappa: **0.6400**, substantial agreement) on a balanced 100-sentence sample.
- Dataset Sizing:** The cleaned dataset contains **3,478 sentence-question records** split into 2,156 training records (15.72% salient) and 1,322 validation records (25.42% salient) across 75 unique contexts (640 QA pairs).

### 3. Rigorous Significance Testing (Welch's T-Test)

- Syntactic Complexity:** Salient sentences exhibit significantly deeper dependency parse trees (mean **6.57** vs. **6.16**,  $p < 0.0001$ ) and larger token dependency distances (mean **3.32** vs. **3.06**,  $p < 0.0001$ ).
- Information Coherence Drop:** An unsupervised GPT-2 sentence deletion coherence drop test proved that removing salient sentences causes a significantly larger paragraph-level surprisal increase than removing non-salient sentences ( $p < 0.05$ ).
- Readability Insignificance:** Flesch Reading Ease and Gunning Fog indices showed no statistically significant differences ( $p > 0.1$ ), proving basic text readability is not a strong saliency discriminator.

### 4. Class Balancing Formulations

To mitigate SQuAD's inherent positional bias (66.37% of answers lie in sentences 0-2), we evaluated 5 training-only balancing methods:

| Balancing Method   | Training Sizing & Class Split      | Formulation & Rationale                                                              |
|--------------------|------------------------------------|--------------------------------------------------------------------------------------|
| None (Baseline)    | 2,156 records (339 Pos, 1,817 Neg) | Natural SQuAD unbalanced distribution (prior class probability $P(Y=1) = 15.72\%$ ). |
| Pairwise (RankNet) | 1,831 pairs (3,662 inputs)         | Trains on difference vectors of context-matched salient/non-salient pairs.           |
| Cluster            | 678 records (339 Pos, 339 Neg)     | Clusters negatives ( $K=339$ ) and selects representative cluster centroids.         |
| RST-Neighborhood   | 678 records (339 Pos, 339 Neg)     | Mines hard negatives positionally and structurally close in the RST tree.            |
| DSNB               | 678 records (339 Pos, 339 Neg)     | Mines hard negatives combining position, SBERT similarity, and RST tree depth.       |

### 5. Comparative Experimental Results

The table below details a representative summary of performance metrics on the natural, unbalanced validation set. It compares the baseline rule-based heuristic, the combined linear classifier, and the proposed Heuristic-Guided BERT across all balancing techniques:

| Model Configuration | Balancing        | Accuracy | F1     | NDCG   | MRR    | MAP    |
|---------------------|------------------|----------|--------|--------|--------|--------|
| Combined LR         | Pairwise         | 0.3759   | 0.4452 | 0.9492 | 0.8924 | 0.8856 |
| Combined LR         | Cluster          | 0.7648   | 0.5499 | 0.9342 | 0.8586 | 0.8476 |
| Combined LR         | RST-Neighborhood | 0.6967   | 0.5774 | 0.9271 | 0.8366 | 0.8301 |
| Combined LR         | DSNB             | 0.7224   | 0.5853 | 0.9195 | 0.8142 | 0.8067 |
| Heur-BERT (RST)     | Pairwise         | 0.7534   | 0.6386 | 0.9561 | 0.9140 | 0.9023 |
| Heur-BERT (RST)     | Cluster          | 0.8101   | 0.6303 | 0.9478 | 0.8783 | 0.8655 |
| Heur-BERT (RST)     | RST-Neighborhood | 0.6982   | 0.5387 | 0.9255 | 0.8269 | 0.8136 |
| Heur-BERT (RST)     | DSNB             | 0.7890   | 0.6225 | 0.9458 | 0.8785 | 0.8653 |

*Calibration Note:* Balanced training (DSNB/Pairwise) shifts the prediction probability scale, causing F1/Accuracy drops at a default 0.5 threshold. However, they achieve excellent ranking metrics (NDCG/MRR) and eliminate SQuAD's positional shortcut. Threshold calibration completely resolves this F1 shift.

### 6. Key Standardized Coefficients (Feature Importance)

Top positive and negative features for the DSNB combined Logistic Regression model:

| Subsystem  | Feature Name    | Coefficient | Interpretation                                                         |
|------------|-----------------|-------------|------------------------------------------------------------------------|
| Linguistic | word_count      | +1.1044     | Positive: Longer sentences contain detailed facts.                     |
| Semantic   | align_sem_sim   | +0.8544     | Positive: Strong SBERT cosine similarity with question.                |
| Discourse  | rel_rst_n_ratio | +0.7203     | Positive: Sentences with primary nucleus (rhetorical) roles.           |
| Linguistic | char_count      | -0.8539     | Negative: Prefers shorter character length given word count (density). |
| Surprisal  | rel_surp_ratio  | -0.6146     | Negative: Low relative surprisal indicates context semantic priming.   |
| Discourse  | rel_rst_depth   | -0.4538     | Negative: Deeply nested subtrees are less salient than main clauses.   |

### 7. Ongoing Work & Next Steps

- **Dataset Scaling:** Scaling the feature extraction pipeline to 500+ contexts to increase pairs to ~25,000+ to evaluate transformer behavior on a larger pool.
- **Threshold Calibration:** Optimizing the prediction decision thresholds on validation splits to improve pointwise F1 scores of balanced models.
- **Downstream Task Integration:** Integrating the trained saliency models into a Question Generation pipeline (e.g., T5) and evaluating BLEU/ROUGE improvements.